

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 31

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Negative • Positive • Negative • Positive	4
01.2	• The response to a change acts to re-establish / return the system to the previous equilibrium • So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback	2
01.3	• eg the possible release of methane hydrate at the end of the Permian era • Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium	2
01.4	• Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) • Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
02.1	• Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding); species higher up food chains need larger areas (lower density) • Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) • Shape: the perimeter has edge conditions (a mix of neighbouring habitats); a compact shape maximises core habitat for species needing core conditions • Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians)	4

Question	Marking guidance	Marks
02.2	- Varied topography: banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) - Islands provide roosting / nesting sites safe from terrestrial predators (foxes) - Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet); scrapes for skylarks; different water depths for waders and ducks	3
02.3	- Mobile species (birds, flying insects) colonise easily - Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced; keystone species especially	2
02.4	Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out; avoids hazards (eg rope bridges for howler monkeys over roads)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	
03.1	- Timber: building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) - Fuel: fuelwood and charcoal - still the main energy source for most people in LEDCs - Food: fruit, nuts, bushmeat, livestock fodder; crops of forest origin (coffee, cacao, bananas, Brazil nuts); pigs and chickens were forest animals - Fibres: paper (wood pulp cellulose), cotton, viscose / rayon - Medicines: quinine (cinchona), taxol (yew), aspirin (willow); most species not yet researched	5
03.2	- Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time; total forest carbon (vegetation plus soil) is about double that in the atmosphere - Growing trees sequester CO₂ by photosynthesis and release oxygen; forest soils hold more carbon than the trees (especially boreal) - Planting / protecting forests removes CO₂ and reduces the enhanced greenhouse effect	3
03.3	- Highest transpiration of any ecosystem increases downwind rainfall; interception increases evaporation; deep organic soils retain water and delay runoff so river flow is more even and flooding reduced - Low albedo absorbs sunlight; heat stored and released at night reduces day-night temperature extremes; canopy microclimate: lower light, wind, higher humidity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	
04.1	- Extensive: low inputs of energy, food and chemicals so low cost and low environmental impact; sustainable; but low yield per area and dependent on natural productivity - Intensive: control of all limiting factors gives very high yield and profit per area; but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes - Judgement depends on species, market and site	4

Question	Marking guidance	Marks
04.2	- Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation - Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily; fry fed fishmeal pellets and moved through tanks as they grow - After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg	3
04.3	- About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection - Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose - Eyestalk ablation stimulates breeding; high densities increase disease so antibiotics / pesticides used	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	
05.1	- Hydrosere - Single-celled algae (and floating plants) - arrive on birds / insects / from soil	2
05.2	- Pioneers and later plants grow and die; dead organic matter plus sediment washed in fill the pond so water becomes shallower - As depth falls rooted plants (pondweeds then emergent reeds with leaves above water) can anchor and colonise; emergents shade submerged plants which die out - Sediment / soil accumulates and becomes able to support larger plants; water-tolerant trees (willow, alder) colonise the waterlogged soil - Transpiration by trees and further sediment dry the soil and make it aerobic so oak can colonise; the denser canopy shades out smaller plants - Each community changes abiotic conditions (depth, light, nutrients, oxygen), making it suitable for the next colonisers which out-compete them; conditions become less extreme and biotic factors increasingly important	5
05.3	- Starting conditions (bare rock vs open water) differ but each stage makes conditions less extreme and builds soil / organic matter - The final community is controlled by the regional climate (temperature, water availability), which is the same for both - climatic climax (eg oak woodland in the UK)	2
05.4	Lichens and algae colonise bare rock (extreme temperatures, no soil, little water); rock fragments and dead organic matter accumulate; mosses, then grasses and ferns as thin soil forms; wind-dispersed trees (birch) then animal-dispersed (oak, beech)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 73-76; AQA spec 3.1.3.3.	
06.1	- Each regional population has its own unique gene pool / DNA profile - DNA from seized timber or ivory is compared with the database of regional populations - This identifies where the material came from, showing whether it was legally harvested and where poaching occurs	3
06.2	- Radio / GPS transmitters use radio waves - work in air and fresh water, giving positions (GPS stores or transmits location) - Radio waves do not travel through seawater so acoustic transmitters (sound) are used for marine animals, giving real-time positions	2

Question	Marking guidance	Marks
06.3	• Drones / ROVs, AUVs (eg Autosub), balloons, aircraft, satellites, animals (tags) • Drones: cheap, low disturbance, high-resolution images of areas hard to reach (cliffs, canopy, ice), repeated surveys; infrared cameras detect animals at night	3
06.4	• Cameras record automatically (visible light by day, infrared at night) without researchers being present • So animals behave naturally and nests are not deserted; monitoring continues 24 hours	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-406; AQA spec 3.7.2.3.	
07.1	• $600 + (90 + 150) - (220 + 80) = 540$ • Stock fell by 60 so the catch exceeds the MSY / unsustainable	2
07.2	• $A + G - M = 90 + 150 - 80 = 160$	1
07.3	• Catch is landed and weighed at ports so it is recorded • Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed • Representative sampling of a large enough proportion of the area is impractical / expensive; survival of eggs and larvae varies greatly between years	3
07.4	• CPUE: if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained; a better measure than catch size which technology can prop up • Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	
08.1	• B	1
08.2	• Rainforest: high light, warm, wet all year, stable conditions for long periods, continuous productivity, many niches (stratification) • Antarctica: very low temperatures, low precipitation, extreme seasonal light, 98% ice cover so almost no terrestrial life • Antarctic biodiversity is marine, supported by upwelling nutrients and the summer bloom; rainforest supports over half of terrestrial species on 10% of land • Both are important carbon stores / climate regulators but rainforest species are specialists vulnerable to small changes; Antarctic species adapted to extreme fluctuation	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
09.1	• Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive • Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light; diversity falls • Hedgerow removal destroys habitat / corridors and nesting sites; loss of natural pest predator habitats increases pest damage; larger fields increase wind erosion; genetic contamination and introduced species also accepted	6
09.2	• Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows • These communities depend on the farming continuing	2
09.3	• CO₂ from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery; ploughing increases soil respiration • Methane from livestock intestines and rice paddies; nitrous oxide from manure and nitrogen fertilisers	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions • Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection • Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE • Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution • Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels • Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population • Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support	9
10.2	• Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Question	Marking guidance	Marks
11.1	Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Species selection: market, range of tolerance, controllability • Abiotic control: temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO₂, topography, relief, wind - methods and examples • Biotic control: pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota • Manipulation of food species: population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM; energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses • Impacts: habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion; positive: plagioclimax habitats • Social, economic and political influences shaping methods • Sustainable strategies and evaluation; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-337; AQA spec 3.5.1.	25

Question	Marking guidance	Marks
11.2	Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Physical features: mass / gravity retaining an atmosphere and giving pressure for liquid water; distance from the Sun and the Goldilocks zone; speed of rotation reducing temperature extremes; tilt producing seasons; magnetic field deflecting the solar wind; the Moon stabilising the orbit and producing tides • Importance of liquid water: solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • Development of life: Archaea at vents; photosynthesis; oxygen reacting with iron then accumulating; ozone layer allowing colonisation of land; aerobic organisms • Life changing conditions: carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases • Methods of studying past conditions: proxy data, isotope ratios, ice cores, Mg:Ca ratios; understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	25